

Student Name:

Student Number:



GOSFORD HIGH SCHOOL

Year 11 Preliminary 2012

Yearly Examination (Task 3)

MATHEMATICS

Time: 2 Hours (*plus 5 minutes reading time*)

General Instructions:

- Board approved calculators may be used.
- Write using black or blue pen.
- All necessary working should be shown in Questions 11 – 16
- Write your student number **AND** name on this examination paper

Total Marks – 70

Section 1 - 10 Marks

- Attempt Questions 1 – 10 on the separate answer sheet. (No working is to be shown on this sheet – use the question sheet.)
- Allow about 15 minutes for this section.

Section 2 - 60 Marks

- Attempt all Questions 11 – 16
- Start each question in a new booklet.
- Allow about 1 hour 45 minutes for this section

This paper **MUST NOT** be removed from the examination room.

SECTION 1: (10 Marks)

- Attempt all questions.
 - Allow about 15 minutes for this section.
 - Use the multiple choice answer sheet (shade in carefully).
 - Any working out can be completed on this question sheet. (**Do not** do working on the answer sheet.)
-

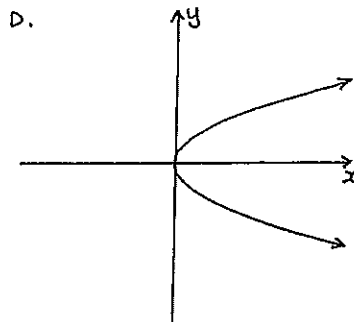
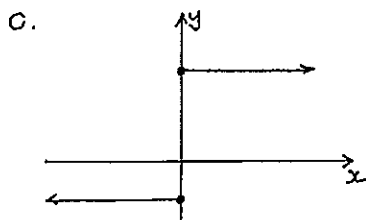
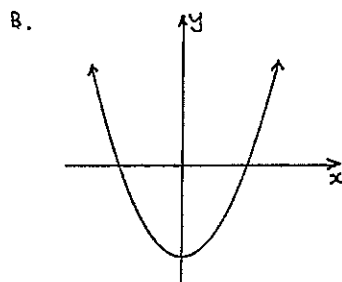
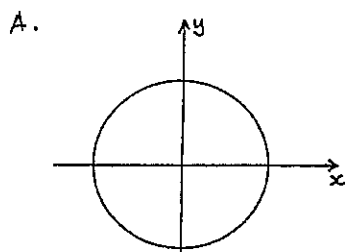
1. What is the value of $\frac{6.4+8.7}{5.9-2.3}$ correct to 2 significant figures?

- A. 4.19
- B. 4.2
- C. 5.57
- D. 5.6

2. $\frac{1+\sqrt{3}}{2-\sqrt{3}} =$

- A. $5 + 3\sqrt{3}$
- B. $\frac{5+3\sqrt{3}}{7}$
- C. $\sqrt{3} - 1$
- D. $\frac{\sqrt{3}-1}{7}$

3. Which of the following graphs represents a function?



4. What is the value(s) of θ , in the domain $0^\circ \leq \theta \leq 360^\circ$, when $\cos \theta = \frac{1}{2}$
- A. 60°
- B. 60° and 120°
- C. 60° and 240°
- D. 60° and 300°

5. Expand and simplify $3(2x - 1) - (x - 4)$

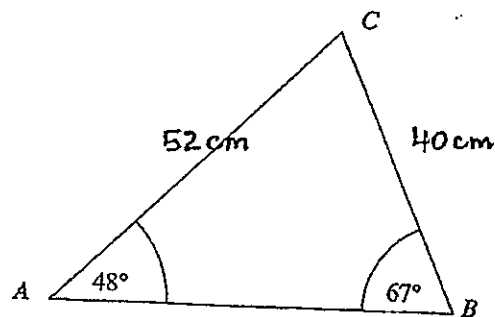
- A. $5x - 7$
- B. $5x - 5$
- C. $5x + 1$
- D. $5x + 3$

6. Which of the following is true for the function

$$f(x) = \frac{3}{x^2} - 2 \text{ ?}$$

- A. Even function
- B. Odd function
- C. Neither odd nor even function
- D. Zero function

7. What is the correct expression for the area of triangle ABC?



Not to scale

- A. $\frac{1}{2} \times 52 \times 40 \times \cos 67^\circ$
- B. $\frac{1}{2} \times 52 \times 40 \times \sin 67^\circ$
- C. $\frac{1}{2} \times 52 \times 40 \times \cos 65^\circ$
- D. $\frac{1}{2} \times 52 \times 40 \times \sin 65^\circ$

8. The domain and range for the function $y = 7 - x^2$ is:

- A. $x \leq 7$, all real y
- B. $x \leq \sqrt{7}$, all real y
- C. All real x , $y \leq 7$
- D. All real x , $y \leq \sqrt{7}$

9. The values of k for which the expression

$$-\frac{x^2}{4} - x - k$$

is a negative definite is:

- A. $k < 1$
- B. $k > 1$
- C. $k < 4$
- D. $k > 4$

10. For the quadratic equation $x^2 + ax + c = 0$, the roots α and β are reciprocals of each other. Which statement **must** be true?

- A. $\alpha \beta = \frac{c}{a}$
- B. $\alpha \beta = -\frac{c}{a}$
- C. $c = 1$
- D. $\frac{c}{a} = 1$

SECTION II

60 marks

Attempt Questions 11 – 16

Allow about 1 hour 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

All necessary working should be shown in every question

Question 11 (10 marks). Use a SEPARATE booklet.

Marks

a. Express $2.\dot{3}\dot{6}\dot{4}$ as a rational number in simplest form. 2

b. Solve:

i. $\frac{5}{x+3} = \frac{3}{x}$ 2

ii. $|2x + 1| = 3x + 2$ 2

c. Solve the following equations simultaneously: 2

$$y = 2x + 4 \text{ and } y = x^2 + 1$$

d. Simplify fully: 2

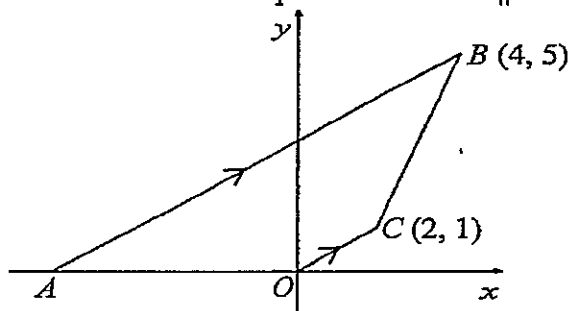
$$\frac{x^2 + y^2 + 2xy}{x^2 - y^2}$$

End of Question 11

Question 12 (10 marks). Use a SEPARATE booklet.

Marks

In the diagram below $ABCO$ is a trapezium with $AB \parallel OC$.



- | | | |
|--------|---|---|
| (i) | Find the coordinates of the midpoint of BC . | 1 |
| (ii) | Calculate the exact length of OC . | 1 |
| (iii) | Find the gradient of OC . | 1 |
| (iv) | Find the size of $\angle AOC$, correct to the nearest degree. | 1 |
| (v) | Show that the equation of the line AB is $x - 2y + 6 = 0$. | 1 |
| (vi) | Find the coordinates of A . | 1 |
| (vii) | Show that the perpendicular distance from C to the line AB is $\frac{6\sqrt{5}}{5}$. | 2 |
| (viii) | Hence, or otherwise, calculate the area of trapezium $ABCO$. | 2 |

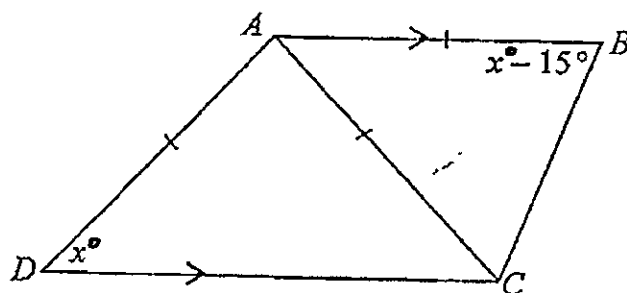
End of Question 12

Question 13 (10 marks). Use a SEPARATE booklet.

Marks

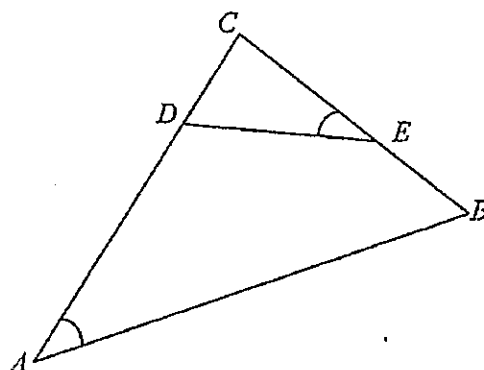
- a. In quadrilateral ABCD, $AB = AC = AD$ and $AB \parallel DC$ as shown
 $\angle ADC = x^\circ$ and $\angle ABC = x^\circ - 15^\circ$

2



Find the value of x giving reasons.

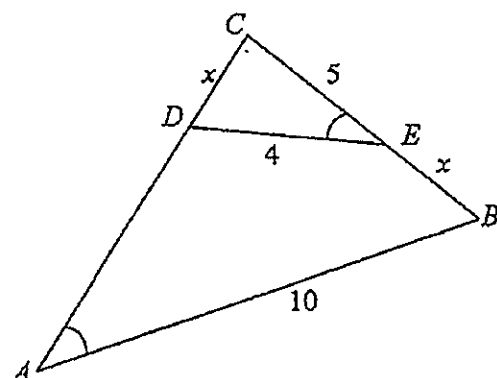
- b. In the diagram ABC is a triangle.
 The points D and E lie on AC and BC respectively so that $\angle BAC = \angle DEC$



Copy or trace the diagram into your writing booklet.

- i. Prove that $\triangle ABC \parallel \triangle EDC$

2



- ii. Hence, if $EC = 5$ cm, $DE = 4$ cm, $AB = 10$ cm and $DC = BE = x$ cm, as shown in the diagram above, find the value of x .

2

Give reasons for your answer.

Question 13 (Continued)

c. c. A function is defined as:

$$f(x) = \begin{cases} x - 5 & \text{for } x \geq 0 \\ -2 & \text{for } -1 < x < 0 \\ 2 + x^2 & \text{for } x \leq -1 \end{cases}$$

- | | | |
|------|---|---|
| i. | Evaluate $f(2) + f(-1)$ | 1 |
| ii. | On a one third page number plane neatly sketch the function $y = f(x)$ clearly indicating all important features. | 2 |
| iii. | Write an expression for $f(a^2)$ | 1 |

End of Question 13

Question 14 (10 marks). Use a SEPARATE writing booklet.

Marks

- a. Prove

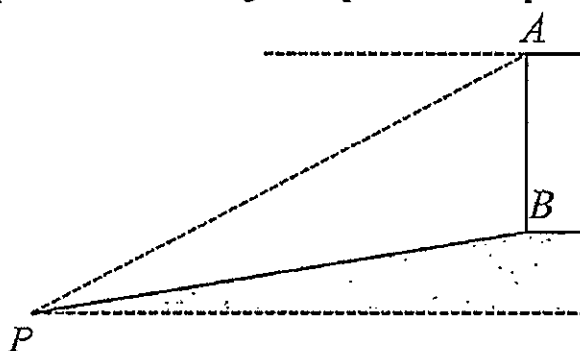
2

$$\frac{1}{\operatorname{cosec}^2 \theta} + \frac{1}{\sec^2 \theta} + \frac{1}{\cot^2 \theta} = \sec^2$$

- b. Given that $\tan \theta = \frac{4}{5}$ and $\sin \theta < 0$, find the exact value of $\cos \theta$

2

- c. A point P is at the foot of the hill which is inclined at 5° to the horizontal. The base B, of a tower AB, is situated 150 metres up the incline of the hill from P. From the top of the tower the angle of depression of the point P is 32° .



- i. Copy the diagram and mark on it the size of $\angle APB$ and of $\angle PAB$.
- ii. Find the height of the tower, correct to the nearest metre.

1

2

- d. Two ships leave a port P at the same time. Ship A travels at 12 km/h on a bearing of 213° . Ship B travels in a south easterly direction at a speed of 8 km/h.

- i. Draw a diagram to illustrate the relative positions of the two ships after 3 hours, correct to the nearest km.
- ii. Calculate the distance between the ships after 3 hours, correct to the nearest km.

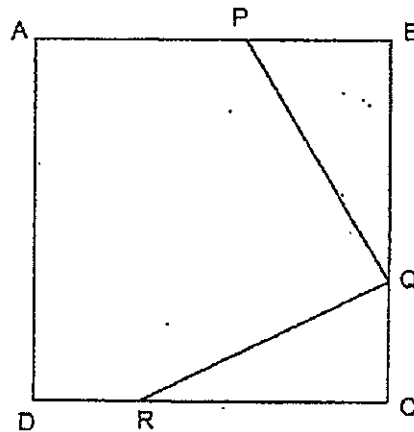
1

2

Question 15 (10 marks). Use a SEPARATE writing booklet.

Marks

- a. In the diagram below, ABCD is a square. P, Q and R are points on sides AB, BC and CD respectively such that $PB = QC = RD$.



Copy the diagram onto your worksheet.

2

Prove that $\triangle BPQ \equiv \triangle CQR$

- b. Solve $x^2 - 5x - 24 \leq 0$

2

- c. If α and β are the roots of the equation $3x^2 + 5x - 4 = 0$, find

i. $\alpha + \beta$ and $\alpha\beta$

1

ii. $\frac{3}{\alpha} + \frac{3}{\beta}$

1

iii. $2\alpha^2 + 2\beta^2$

2

- d. Given the equation $(k + 1)x^2 + (k + 2)x + k = 0$ find the value of k if the equation has equal roots.

2

End of question 15

Question 16 (10 marks). Use a SEPARATE writing booklet.

Marks

- a. Solve for θ , where $-180^\circ \leq \theta \leq 180^\circ$ 2

$$\sqrt{3} \tan \theta + 1 = 0$$

- b. Find the values of a, b and c if 3

$$\frac{x^2 + 1}{x^2 + 3x + 2} \equiv \frac{a}{x + 2} + \frac{bx + c}{x + 1}$$

- c. Solve: 3

$$\frac{1}{x^4} - \frac{3}{x^2} - 4 = 0$$

- d. If $f(x) = x^2 - 3x + 5$, 2

Find an expression, in its simplest form for

$$\frac{f(x + h) - f(x)}{h}$$

End of Examination

SECTION 1.

1/ $4.194444... \text{ (calc)}$

$= 4.2$ B

2/ $\frac{1+\sqrt{3}}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}}$

$= \frac{2+\sqrt{3}+2\sqrt{3}+3}{4-3}$

$= 5+3\sqrt{3}$ A

3/ B

4/ $\cos \theta = \frac{1}{2}$

$\therefore \theta = 60^\circ, 300^\circ$ D

5/ $3(2x-1) - (x-4)$

$= 6x-3-x+4$

$= 5x+1$ C

6/ Since $f(x) = f(-x)$

$\therefore$ even function A

7/ Using the sin rule:

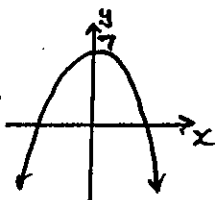
$\frac{42}{\sin 40^\circ} = \frac{22}{\sin 55^\circ}$

$\therefore 42 = \frac{22 \sin 60^\circ}{\sin 55^\circ}$ B

8/ From the graph:

D: all real x

R: $y \leq 7$ C



9/ to be a negative definite

$a < 0$ and $\Delta < 0$

$\Delta = b^2 - 4ac$

$= 1 - 4(-\frac{1}{4})(-k)$

$= 1 - k$

$\therefore 1 - k < 0$

$1 < k$

B

10/ If reciprocals then product of roots = 1

$\therefore c = 1$ C

SECTION II

Q11/ a. let $x = 0.36\dot{4}$

$\therefore 100x = 36.46464...$

$x = 0.36464...$

$99x = 36.1$

$\therefore 990x = 361$

$x = \frac{361}{990}$

$\therefore$ answer = $2 \frac{361}{990}$

b. (1) $\frac{5}{x+3} = \frac{3}{x}$

$5x = 3x+9$

$2x = 9$

$x = \frac{9}{2}$

$$(11) |2x+1| = 3x+2$$



$$2x+1 = 3x+2 \quad \text{OR} \quad 2x+1 = -(3x+2)$$

$$-1 = x$$

$$2x+1 = -3x-2$$

$$5x = -3$$

$$x = -\frac{3}{5}$$

check solutions :

$x = -1$ ✗ does not work

$x = -\frac{3}{5}$ ✓ works

$\therefore x = -\frac{3}{5}$ is the only solution

$$c. \quad y = 2x + 4 \quad - (1)$$

$$y = x^2 + 1 \quad - (2)$$

Sub (1) in (2) :

$$2x+4 = x^2+1$$

$$0 = x^2 - 2x - 3$$

$$0 = (x-3)(x+1)$$

$$\therefore x = 3, -1$$

$$y = 10, 2$$

$$d. \quad \frac{x^2 + y^2 + 2xy}{x^2 - y^2}$$

$$= \frac{(x+y)^2}{(x-y)(x+y)}$$

$$= \frac{x+y}{x-y}$$

Q12 (i) Midpt BC

$$= \left(\frac{4+2}{2}, \frac{5+1}{2} \right)$$

$$= (3, 3)$$

$$(ii) \quad OC^2 = 2^2 + 1^2$$

$$= 5$$

$$\therefore OC = \sqrt{5} \text{ units}$$

$$(iii) \quad m_{OC} = \frac{1-0}{2-0} = \frac{1}{2}$$

$$(iv) \quad \text{let } \angle COx = \theta$$

$$\therefore \tan \theta = \frac{1}{2}$$

$$\therefore \theta = 26^\circ 34'$$

$$\therefore \angle AOC = 180^\circ - 26^\circ 34'$$

$$= 153^\circ 26'$$

$$= 153^\circ \text{ (nearest deg)}$$

$$(v) \quad M_{AB} = \frac{1}{2}$$

$$\therefore \text{eqn} \Rightarrow y-5 = \frac{1}{2}(x-4)$$

$$2y-10 = x-4$$

$$x-2y+6=0$$

as req.

$$(vi) \quad \text{Sub } y=0$$

$$\therefore x = -6$$

$$\therefore A = (-6, 0)$$

$$(vii) \quad \perp d = \frac{|(1)(2) + (-2)(1) + 6|}{\sqrt{1^2 + (-2)^2}}$$

$$= \frac{6}{\sqrt{5}} = \frac{6\sqrt{5}}{5} \text{ units}$$

as req.

$$(viii) A = \frac{1}{2}(a+b)$$

$$\begin{aligned} \text{Now } d_{AB} &= \sqrt{(4+6)^2 + (5-0)^2} \\ &= \sqrt{10^2 + 5^2} \\ &= \sqrt{125} \\ &= 5\sqrt{5} \end{aligned}$$

$$\begin{aligned} \therefore A &= \frac{6\sqrt{5}}{10} (5 + 5\sqrt{5}) \\ &= \frac{(6\sqrt{5})^2}{10} \\ &= 18 \text{ units}^2 \end{aligned}$$

13/ a. $\angle ACD = x^\circ$ (equal base $\angle$'s in isosceles Δ)
 $\angle CAB = x^\circ$ (alternate $\angle$'s in parallel lines are equal)

Also $\angle ACB = x^\circ - 15^\circ$ (equal base $\angle$'s in isosceles Δ)

$$\therefore x^\circ + x^\circ - 15^\circ + x^\circ - 15^\circ = 180^\circ$$

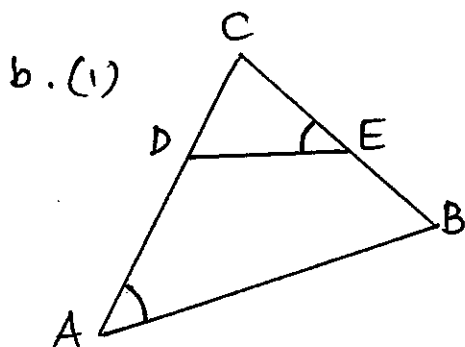
($\angle$ sum of $\Delta = 180^\circ$)

$$3x^\circ - 30^\circ = 180^\circ$$

$$3x^\circ = 210^\circ$$

$$x^\circ = 70^\circ$$

$$\therefore x = 70$$



$$\angle BAC = \angle DEC \text{ (given)}$$

$$\angle BCA = \angle DCE \text{ (common)}$$

$$\therefore \Delta ABC \parallel \Delta EDC \text{ (equiangular)}$$

(ii) Since similar, corresponding sides are in proportion

$$\text{i.e. } \frac{CD}{CB} = \frac{DE}{AB}$$

$$\frac{x}{5+x} = \frac{4}{10}$$

$$10x = 20 + 4x$$

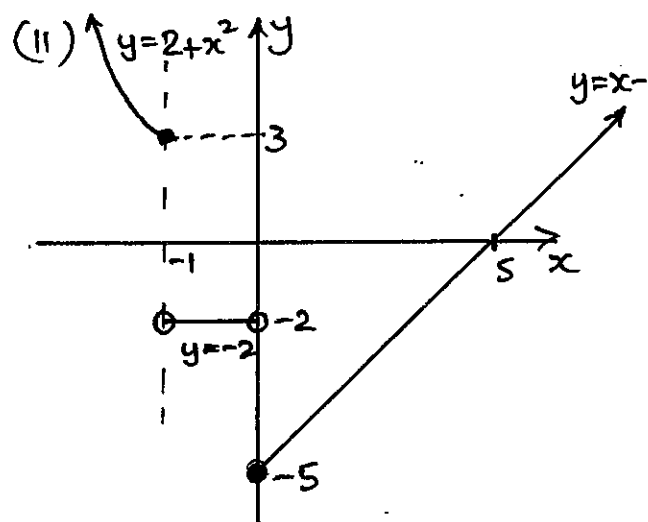
$$6x = 20$$

$$x = \frac{10}{3}$$

$$c. (i) f(2) = 2 - 5 = -3$$

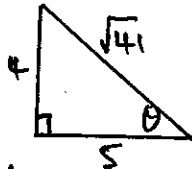
$$f(-1) = 2 + (-1)^2 = 3$$

$$\therefore f(2) + f(-1) = -3 + 3 = 0$$



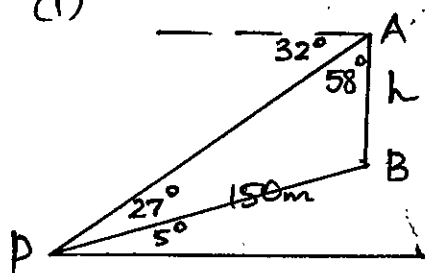
$$(iii) f(a^2) = a^2 - 5$$

14 a. $\text{LHS} = \sin^2 \theta + \cos^2 \theta + \tan^2 \theta$
 $= 1 + \tan^2 \theta$
 $= \sec^2 \theta$
 $= \text{RHS as req.}$

b. $\tan \theta = \frac{4}{5} \Rightarrow$ 
 and $\sin \theta < 0$
 $\therefore \theta$ is in 3rd Quad.

$\therefore \cos \theta = -\frac{5}{\sqrt{41}}$

c. (i)

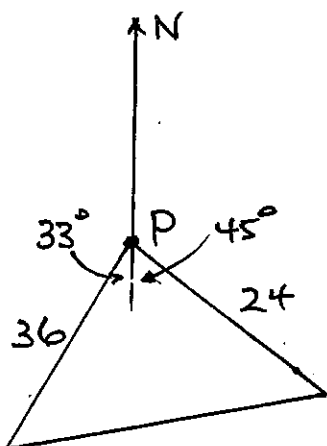


(ii) Using the sin rule

$$\frac{h}{\sin 27^\circ} = \frac{150}{\sin 58^\circ}$$

$$h = \frac{150 \sin 27^\circ}{\sin 58^\circ}$$

d. (i)



(ii) $d^2 = 36^2 + 24^2 - 2 \times 36 \times 24 \times \cos 78^\circ$
 $= 1512.728598 \text{ (calc)}$
 $\therefore d = 38.8938 \dots \text{ (calc)}$
 $\therefore \text{distance is } 39 \text{ kms.}$

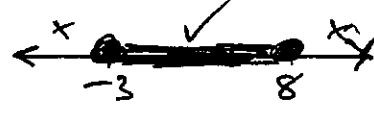
Q15 a. $\angle PBQ = \angle RCQ$ ($90^\circ \angle$ in a square)

$PB = QC$ (given)

$BQ = RC$ (equal lengths subtracted from equal sides of a square)

$\therefore \triangle BPQ \equiv \triangle CQR$ (SAS)

b. Consider $x^2 - 5x - 24 = 0$
 $(x-8)(x+3) = 0$
 $x = 8, -3$

Test in $x^2 - 5x - 24 \leq 0$ 

$$\therefore -3 \leq x \leq 8$$

c. (i) $\alpha\beta = \frac{c}{a} = -\frac{4}{3}$

$$\alpha + \beta = -\frac{b}{a} = -\frac{5}{3}$$

(iii) $2\alpha^2 + 2\beta^2 = 2(\alpha^2 + \beta^2)$
 $= 2[(\alpha + \beta)^2 - 2\alpha\beta]$
 $= 2\left[\left(-\frac{5}{3}\right)^2 - 2\left(-\frac{4}{3}\right)\right]$
 $= 2\left(\frac{25}{9} + \frac{8}{3}\right)$
 $= \frac{98}{9}$

$$\begin{aligned}
 \text{(11)} \quad \frac{3}{\alpha} + \frac{3}{\beta} &= \frac{3\alpha + 3\beta}{\alpha\beta} \\
 &= \frac{3(\alpha + \beta)}{\alpha\beta} \\
 &= \frac{3\left(-\frac{5}{3}\right)}{-\frac{4}{3}} \\
 &= \frac{15}{4}
 \end{aligned}$$

d. Equal roots when $\Delta = 0$

$$\text{i.e. } (k+2)^2 - 4(k+1)k = 0$$

$$k^2 + 4k + 4 - 4k^2 - 4k = 0$$

$$-3k^2 + 4 = 0$$

$$4 = 3k^2$$

$$\frac{4}{3} = k^2$$

$$\pm \frac{2}{\sqrt{3}} = k$$

Q16 a. $\sqrt{3}\tan\theta + 1 = 0$

$$\sqrt{3}\tan\theta = -1$$

$$\tan\theta = -\frac{1}{\sqrt{3}}$$

for $0^\circ \leq \theta \leq 360^\circ$

$$\theta = 150^\circ, 330^\circ$$

So in $-180^\circ \leq \theta \leq 180^\circ$

$$\theta = -30^\circ, 150^\circ$$

$$b. \frac{x^2+1}{x^2+3x+2} \equiv \frac{a}{x+2} + \frac{bx+c}{x+1}$$

$$\begin{aligned}
 \therefore x^2+1 &\equiv a(x+1) + (bx+c)(x+2) \\
 &= ax+a+bx^2+2bx+cx+2c \\
 &= bx^2 + (a+2b+c)x + (a+2c)
 \end{aligned}$$

$$\therefore b=1 \quad -\textcircled{1}$$

$$a+2b+c=0 \quad -\textcircled{2}$$

$$a+2c=1 \quad -\textcircled{3}$$

$$\textcircled{2} \Rightarrow a+c=-2 \quad -\textcircled{4}$$

$$\textcircled{3} \Rightarrow a+2c=1 \quad -\textcircled{5}$$

$$\textcircled{4} - \textcircled{5} \quad -c = -3$$

$$c=3$$

$$\therefore a=-5$$

$$\therefore a=-5, b=1, c=3$$

c. let $A = \frac{1}{x^2}$

$$\therefore A^2 - 3A - 4 = 0$$

$$(A-4)(A+1) = 0$$

$$\therefore A = 4, -1$$

$$\therefore \frac{1}{x^2} = 4 \text{ or } \frac{1}{x^2} = -1$$

$$\therefore x^2 = \frac{1}{4}$$

$$x = \pm \frac{1}{2} \text{ are}$$

the only solutions.

no real
solution

$$d. f(x) = x^2 - 3x + 5$$

$$\begin{aligned} f(x+h) &= (x+h)^2 - 3(x+h) + 5 \\ &= x^2 + 2xh + h^2 - 3x - 3h + 5 \end{aligned}$$

$$\therefore \frac{f(x+h) - f(x)}{h}$$

$$= \frac{x^2 + 2xh + h^2 - 3x - 3h + 5 - x^2 + 3x - 5}{h}$$

$$= \frac{2xh + h^2 - 3h}{h}$$

$$= 2x + h - 3$$